Figure 9.8 shows a pre-constructed k-d tree that we are going to use as a starting point for this section. In order to focus on the details of the algorithm, we are going to use a simplified view, showing points on a cartesian plane where axes are omitted. Points are just denoted with capital letters, we are leaving out city names for now, to abstract out the context and focus on the algorithms. Vertical (red) and horizontal (blue) splits are still shown, while instead we won't highlight the regions as we have done in figure 9.6 etc...; as a consequence, we can fill tree nodes (instead of edges) with red or blue to show that vertical or horizontal splits will be used at a certain level.

While in figure 9.8 coordinates are shown next to both nodes and points, we might sometimes omit them in the next figures, for the sake of cleanness.

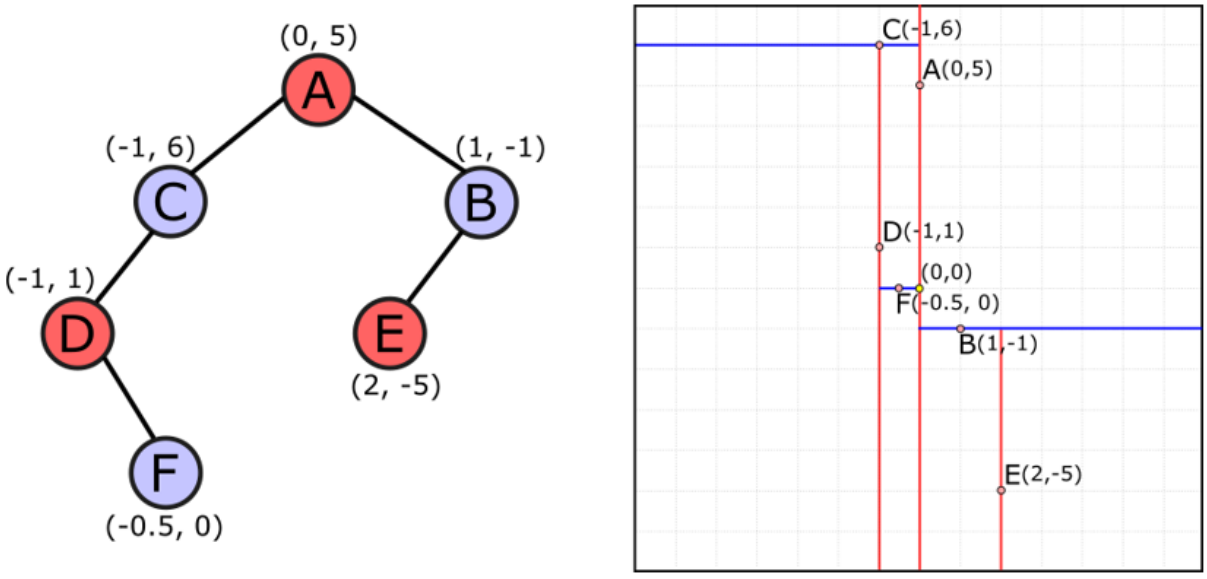


Figure 9.8 An example k-d tree for a 2-D dataset. On the left, the tree representation. On the right, a